

Marine biology study: the hydrothermal vent ecosystem. Researchers studying the Mid-Atlantic Ridge have documented a dense food web that depends entirely on chemosynthesis rather than sunlight. The central organism is *Riftia pachyptila*, a giant tubeworm that lacks a mouth and gut. *Riftia pachyptila* hosts symbiotic bacteria inside a specialized organ called the trophosome. The bacteria, which belong to the genus *Candidatus Endoriftia*, oxidize hydrogen sulfide to produce organic carbon. This process sustains the vent community.

The vent fields also contain the blind shrimp *Rimicaris exoculata*. *Rimicaris exoculata* grazes on bacterial mats near the chimney walls and harbors bacteria on its dorsal shell. Both *Riftia pachyptila* and *Rimicaris exoculata* rely on hydrogen sulfide produced by the vent fluid. The vent fluid reaches temperatures above 350 degrees Celsius. Nearby, the mussel *Bathymodiolus thermophilus* filters particulate organic matter from the cooler water surrounding the vents.

Vent chemistry shapes the whole community. The chimney structures are built from sulfide minerals, principally pyrite and chalcopyrite, deposited as hot fluid cools. Vesicles in the chimney release the hydrogen sulfide that the Endoriftia bacteria require. Because sunlight never penetrates this habitat, the food web of *Riftia pachyptila*, *Rimicaris exoculata*, and *Bathymodiolus thermophilus* is entirely chemotrophic. Scientists continue to survey new vent fields to expand the known range of Endoriftia and its bacterial mats.